Vector Spaces I–II: Subspaces, Span, Independence, Basis and Dimension: Answer Key

Problem sheet for the Day 3 classes · CSIR NET / GATE previous year questions

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	(C)	(C)	(C)	(A)	(B)	(B)	(A)	(C)	(C),(D)	(A)

Hints and short solutions

1. **(C)** GATE 2016 $(x+y)-(y+z)=x-z$, so P fails. For Q the coefficient matrix has determinant $-6\neq 0$.
2. **(C)** CSIR NET Dec 2019 7 anti-diagonals, each one condition on its own entries; the conditions are independent: $16-7=9$.
3. **(C)** CSIR NET Dec 2016 Limits are linear, so S_1, S_3 are closed under sums and scalars; S_2 does not contain 0.
4. **(A)** CSIR NET June 2016 The normal to the plane is $x\times y=(\delta_1,-\delta_2,\delta_3)$; equivalently $\det[(u,v,w);x;y]=0$.
5. **(B)** CSIR NET June 2017 Two independent equations: $\dim N(A)=2$, so rank $=4-2=2$. The vectors in (D) do not satisfy $x+y+z=0$.
6. **(B)** CSIR NET June 2017 $\{(x-x_0)^2,\ x-x_0,\ 1\}$ is a basis of the polynomials of degree ≤ 2: coordinates exist and are unique (Taylor at x_0).
7. **(A)** CSIR NET Dec 2014 Row reduction gives the RREF rows $(1,2,0)$ and $(0,0,1)$.
8. **(C)** CSIR NET June 2019 If $a\sin x+b\cos x+c\tan x=0$ on $[0,1]$: $x=0$ gives $b=0$; then $\sin x\,(a+c\sec x)=0$ forces $a=c=0$ since $\sec x$ is not constant.
9. **(C),(D)** CSIR NET Dec 2016 V_1 misses 0; in V_2, $E_{11}+(I-E_{11})=I$. Trace is linear; V_4 is the image of the linear map $A\mapsto BA$.
10. **(A)** CSIR NET June 2019 $\deg q=n+1$, $\deg r=n-1$, both non-zero: independent. A combination $aq+br$ has degree $n+1$ (if $a\neq 0$) or $\leq n-1$, never n. (D) fails for $p=1+x$.